Ezra Ornstein

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Skills

Tools: Altium, LTspice, Oscilloscope, Digital Multimeter, Soldering, Harnessing, SOLIDWORKS, AutoCAD, JLCPCB
Languages and Protocols: C, C++, PWM, DShot, I2C, SPI, UART, CAN, USB PD

Experience

Electrical Engineering Co-op – Waterloo Aerial Robotics Group May 2026 – Aug 2026
University of Waterloo

- Designed, simulated, assembled, and programmed a USB-C PD sink to give the firmware team a bench-safe power source for testing, eliminating Li-ion dependency so the team could run continuous testing without battery swaps
- Designed, simulated, assembled, and validated a high-efficiency buck converter PCB to step the UAV battery down to 12V to power drone peripherals
- Assisted electrical team members with schematic and layout reviews, board assembly and testing, and component requisitions, ensuring boards were designed and assembled correctly and efficiently

Engineering Projects

USB-C PD Sink With PPS Control — *Altium, LTspice, Oscilloscope, DMM, Soldering, C* Jun 2026 – Jul 2026

- Designed, verified, assembled and tested a USB-C PD sink board supporting PPS negotiation via an ESP32-C3-MINI MCU communicating with an AP33772S USB PD IC over I2C to power systems and boards during firmware testing, replacing the need for Lion batteries which must be recharged between uses
- Integrated user-facing controls (buttons, potentiometer) and an onboard OLED display for real-time setpoint and measured voltage/current feedback, allowing simple use and understanding for the firmware team
- Programmed ESP32 firmware and web server connected over Wi-Fi to phones and laptops for in-depth readout
- Implemented four configurable 5V-20V at up to 4A power outputs, and fixed 3V3 header at up to 1A to support diverse downstream loads, allowing the firmware team to test any system they need with one board

Custom FPV Flight Controller — *Altium, LTspice, Soldering, Digital Multimeter, Oscilloscope* Ongoing

- Designed a 30.5×30.5mm flight controller around an STM32F722RET6 MCU with onboard BMP390 barometer and ICM42688-P IMU, verifying full component compatibility for Betaflight integration
- Integrated multiple SPI, I2C and UART connections, a USB-C port with differential routing to power the board and flash the MCU, and added connectors and pads for modular drone assembly
- Integrated multistage power system with LMR33630 buck converters from the battery to 5V and 9V, an LM66200 Or-ing diode to combine battery and USB 5V, and an AP7361EA 3V3 LDO to power the MCU and peripherals

VTOL UAV Wiring Architecture — *Altium, SOLIDWORKS, Harnessing, Soldering, DMM* Ongoing

- Designed complete power and signal harnessing for a 2.5m wingspan, 5-motor VTOL UAV (4 lift motors + 1 pusher, dual 6S3P battery packs, Pixhawk flight controller)
- Engineered per-wing harness architecture including dual XT60 power connections, 2 ESCs, 5 servo connectors, and a CAN bus network integrating barometer, gyroscope, and GPS sensors
- Produced full interconnect schematics defining signal routing, connector placement, and physical harnessing

Electric Go-Kart Conversion — *Harnessing, Soldering, Digital Multimeter* Aug 2026 – Sep 2026

- Converted a gas-powered (lawnmower-engine) go-kart to full electric drive, redesigning the powertrain and control systems and acheiving 1800W of power at 48V with up to an hour of runtime
- Installed and harnessed new 48V 1800W BLDC motor, motor controller, 12S3P battery, ignition key, forward/reverse and 3-speed switch controls for variable drive modes
- Integrated hydraulic brakes and accelerator pedal assembly interfaced with the motor controller's throttle input for fast and intuitive braking and acceleration

High Efficiency 48V - 12V Buck Converter — *Altium, LTspice, Oscilloscope* May 2026 – Jun 2026

- Designed, simulated, assembled, and validated a 30.5 × 30.5 mm high-efficiency buck converter PCB to step 12S UAV battery voltage (36–50.4 V) down to 12 V, achieving 95% peak efficiency

Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science in Mechatronics Engineering *Expected 2030*